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Primary Hip

Patient Factors That Matter in Predicting Hip Arthroplasty Outcomes: A Machine-Learning Approach



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ABSTRACT

Background: Despite the success of total hip arthroplasty (THA), approximately 10%–15% of patients will be dissatisfied with their outcome. Identifying patients at risk of not achieving meaningful gains postoperatively is critical to pre-surgical counseling and clinical decision support. Machine learning has shown promise in creating predictive models. This study used a machine-learning model to identify patient-specific variables that predict the postoperative functional outcome in THA.

Methods: A prospective longitudinal cohort of 160 consecutive patients undergoing total hip replacement for the treatment of degenerative arthritis completed self-reported measures preoperatively and at 3 months postoperatively. Using four types of independent variables (patient demographics, patient-reported health, cognitive appraisal processes and surgical approach), a machine-learning model utilizing Least Absolute Shrinkage Selection Operator (LASSO) was constructed to predict postoperative Hip Disability and Osteoarthritis Outcome Score (HOOS) at 3 months.

Results: The most predictive independent variables of postoperative HOOS were cognitive appraisal processes. Variables that predicted a worse HOOS consisted of frequent thoughts of work ($\beta = -0.34$), frequent comparison to healthier peers ($\beta = -0.26$), increased body mass index ($\beta = -0.17$), increased medical comorbidities ($\beta = -0.19$), and the anterior surgical approach ($\beta = -0.15$). Variables that predicted a better HOOS consisted of employment at the time of surgery ($\beta = 0.17$), and thoughts related to family interaction ($\beta = 0.12$), trying not to complain ($\beta = 0.13$), and helping others ($\beta = 0.22$).

Conclusions: This clinical prediction model in THA revealed that the factors most predictive of outcome were cognitive appraisal processes, demonstrating their importance to outcome-based research.

Level of evidence: Prognostic Level 1.

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Total hip arthroplasty (THA) offers pain relief, restoration of function, and improved quality of life (QOL) to patients with advanced arthritis [1]. Despite its success, approximately 10%–15% of patients will be dissatisfied with their outcome [2–5]. With the lifetime probability of undergoing THA expected to double by 2040 [6–9], it is critical to understand how we can better meet patient expectations.

Traditionally the success of orthopedic procedures has been measured by surgeon-determined clinical parameters and a lack of complications [10,11]. Recently, there has been a shift toward a more patient-centered analysis, using both disease-specific and generic patient-reported outcome measures (PROMs) to measure pain, function, and health-related quality of life (HRQOL) [12,13]. These are tools that transform patient symptoms and cognitive processes into numerical scores that help provide a complete picture of the patient experience before and after surgery.

While PROMs have provided an abundance of new clinical information to consider, our current understanding of why some patients are satisfied and others are not satisfied continues to be incomplete [14]. Despite what would appear to a surgeon as a successful THA based on radiologic and clinical parameters, a mismatch between the expected and observed patient-reported outcome can still occur [15]. Previous studies in the spine literature have identified this phenomenon and attributed it to response shift, missing important information in our measurements of outcome [16–19]. Efforts to identify specific patient characteristics that are important to arthroplasty outcome have led to conflicting results in regards to age, gender, body mass index (BMI) and preoperative PROMs [3,10,11,20–27]. Even a group of fellowship-trained surgeons were unable to accurately predict the patients undergoing total joint arthroplasty who would achieve a clinically significant improvement in their PROMs postoperatively [28]. This can cloud the shared decision-making process, in which clinicians and patients share evidence and preferences to support an ultimate treatment decision [29].

Artificial intelligence, in particular machine learning, has become increasingly popular in orthopedic research [30], especially when using large data sets to make predictions [31–34]. Machine learning utilizes computer algorithms to learn complex relationships or patterns from empirical data to produce models that link multiple variables to a target variable of interest [35]. More specifically, supervised machine learning identifies relationships between input data (ie, patient demographics, surgical factors, PROMs) and output data (patient outcome), allowing for outcome predictions based on input data [36]. Models have been successfully developed for predicting mortality, readmissions, complications, and length of stay [31–34]. The few attempts utilizing supervised machine learning to predict postoperative PROMs in the total joint arthroplasty population have been retrospective in nature and premised on the binary outcome of whether patients reach a particular minimum clinically important difference in their postoperative PROM [20,37].

The purpose of this study is to explore the value of utilizing supervised machine learning to predict postoperative functional outcomes in THA. This is the first study of its kind. Utilizing patient-derived (demographics, generic and hip-specific PROMs) and surgeon-derived data (surgical approach) from a prospective patient cohort, we attempt to outline the variables that predict the postoperative outcome (Hip Disability and Osteoarthritis Outcome Score [HOOS] [38]). Accurate predictions of the postoperative outcome can help preoperatively to identify patients at risk for dissatisfaction, allowing for targeted individualized counseling and education during the shared decision-making process. A successfully shared decision-making process has been shown to align surgeon and patient expectations, improve patient satisfaction, and ultimately lead to improved outcomes [39].

Methods

Sample and Design

This prospective longitudinal cohort study included adults who were consecutively recruited from an active hip arthroplasty practice at a Canadian academic teaching hospital. The practice includes two direct anterior approach (DAA) surgeons, one anterolateral approach surgeon, and three direct lateral approach surgeons. Patients included were over the age of 18 with a primary diagnosis of osteoarthritis. Patients were excluded if they had certain other relevant diagnoses (ie, osteonecrosis, cancer, acute fracture, inflammatory arthritis), were on immunosuppressant medications, had undergone a previous total joint arthroplasty (TJA), did not provide informed consent, or were unable to understand and complete the English survey-related documents. All patients provided written informed consent prior to completing any questionnaires. Data were collected online or by mail presurgery and again at approximately 3 months postsurgery using a secure Health Information Portability and Accountability Act (HIPAA)-compliant interface [40]. The study was reviewed and approved by the institution's Research Ethics Board.

Measures

Self-Report Measures

We focused on predicting the dependent variable Hip Disability and Osteoarthritis Outcome Score (HOOS) since it is the primary outcome in total hip arthroplasty (THA) [38]. The HOOS Jr. contains two questions about pain and four questions about daily-living function. We included these plus an additional 13 questions about daily-living function from the HOOS long-form, based on pretesting with THA patients who indicated that the additional 13 questions captured important information about their symptom experience. The summary score was computed using all 19 items, [as the mean of the individual items, divided by 4 (the highest possible score for a single answer option), and multiplied by 100 as a scaling factor]. This result was then subtracted from 100 so that higher scores would reflect better outcomes.

Four types of independent variables were considered for inclusion in the analyses described below: (a) clinical and demographic variables; (b) patient-reported health and health habits; (c) cognitive appraisal processes; (d) surgical approach. The *clinical and demographic* candidate predictors included age, gender, race, Hispanic ethnicity, marital status, level of education (less than a college degree, college degree, or advanced degree), employment status (currently working vs not), number of hours worked per week, planned legal action, and worker's compensation status. *Patient-reported health & health habits* variables included HOOS at baseline (97% of values were collected 0–2 months presurgery), smoking (have smoked vs never smoked), body mass index (BMI), and the number of days per week of mild, moderate, and strenuous exercise. Additionally, the *Self-Administered Comorbidity Questionnaire* [41] provided information on whether the patient had each of 14 common comorbidities, and it afforded a count of the number endorsed. This computation excluded arthritis because virtually the entire sample reported having arthritis, an underlying condition leading to the need for THA. *Cognitive appraisal processes* were assessed using the Brief Appraisal Inventory© [42]. This 23-item rating-scale measure helps to describe individual differences in what respondents think about when completing a QOL measure. Finally, the *surgical approach* included direct lateral, anterolateral, and direct anterior methods.

Statistical Analysis

Patients were selected for analysis if they (a) had completed the HOOS items both at baseline and the later follow-up after surgery,

and (b) had provided a set of Appraisal responses at baseline. We ultimately built a final regression model predicting HOOS at about three months postsurgery (mean = 126 days, SD = 119).

Predictive Models

Overall Approach

Our chief aim was to create an accurate predictive model. Even so, in selecting predictors, we incorporated some consideration of cause-and-effect relationships. To this end, we conducted variable selection for candidate predictors in stages—demographic, clinical, patient-reported outcomes, and health habits in the first, and appraisal items in the second. Following classic principles of causal modeling [43], we retained, and in the ultimate model controlled for, predictors chosen in the first stage. Factors that were set in motion earlier, were measured more objectively, and were expected to be more generative, regardless of how their coefficients might have been altered by the inclusion of later predictors. We also conducted bootstrapping and cross-validation in ways designed to extract the most information from our fairly small sample.

Candidate Predictors

A large set of variables were considered as possible predictors, representing the abovementioned four types: (a) clinical and demographic; (b) patient-reported health and health habits; (c) cognitive appraisal processes. Finally, (d) surgical approach was included via forced entry at the end of variable selection.

Variable Selection

Our primary variable selection tool was the Least Absolute Shrinkage Selection Operator or LASSO [44,45]. This popular machine-learning algorithm can reduce overfitting through “penalization.” This involves shrinking regression coefficients, perhaps all the way to zero (which has the effect of excluding a predictor entirely), in order to maximize out-of-sample prediction accuracy. Specifically, LASSO shrinks coefficients to levels that collectively will minimize mean squared error (MSE) in (in our case 25-fold) cross-validated regression. LASSO provides for a tuning parameter, λ , which adjusts the degree of penalization; there is often a trade-off between reducing MSE and creating the most informative model for exploratory purposes. We ultimately chose a level of penalization close to but slightly less than that, which would have strictly minimized MSE.

LASSO selects predictors in a fashion complex enough that it is not applicable to assign them P values. The primary R package used to implement LASS [45] does not report P values; it does not test variables against some null hypothesis but rather constructs models that will minimize prediction error. For these reasons, we do not report P values for our models, although we do report other indicators of stability of estimates.

Predictors Excluded Initially

Variable selection needed to take into account the time-frame of a candidate-predictor (those included in the analysis had to have data collected before the chosen outcome); its distribution; and in one case, its degree of collinearity with others. Appraisal responses collected near the same time as the outcome were excluded. Among demographic and clinical variables, we left out of the LASSO procedure 11 that had overly narrow distributions (ie, dichotomous variables with nearly all responses in one category): arthritis, back pain, asthma, cancer, diabetes, heart disease, kidney disease, liver disease, lung disease, stroke, and stomach ulcer. Hispanic ethnicity, legal action, and workers' compensation also were excluded from variable selection due to few endorsements, and race, due to few

nonwhite patients. Finally, we excluded baseline HOOS as it was collinear with many stage-1 predictors of greater interest; its r with HOOS at 3 months was 0.22.

Treatment of Missing Data

Variable selection using LASSO, and subsequent predictive modeling, used listwise deletion (complete-case analysis). After the exclusions listed above, with the small degree of remaining missingness noted ($\leq 5\%$ per predictor), and with no reason to believe the data were anything but missing-at-random [46], we expected our approach to yield results largely unbiased by missing-data patterns.

Bootstrapped Coefficient Estimates

After checking regression diagnostics and obtaining full-sample coefficient estimates for our best solution, we conducted bootstrapping of regression coefficient standard errors. This entailed 1000 simulation trials, each trial using a randomly selected 90% of patient records to assign all chosen predictors coefficients that would build the best-fitting model for that subsample of data. In each iteration, we preserved the obtained coefficients, and with the simulation complete, we obtained standard deviations for each (“SD_{boot}”) as an estimate of stability and as an alternate to each full-sample standard error.

Cross-Validation

Separately from the operations described above, we conducted cross-validation to see how well the model would perform in predicting HOOS for out-of-sample data, ie, data not included in building the model. We created 1000 splits into a training set (a randomly selected 67% of patient records) and a test set (the remaining 33% in each iteration). Each time the simulation assigned the best estimates of coefficients for that training set, and we noted each resulting solution's R^2 when applied to the test set.

Software

Data were analyzed using IBM SPSS version 26 [47] and the R software [48], including the specialized packages glmnet [45] and plotmo [49].

Results

Sample

Our sample included 160 patients, 56% of whom were male. The sample had a mean age of 67 and an average of 2.2 comorbidities (range 0–5). Cronbach's alpha for the HOOS in this sample was 0.96. Table 1 summarizes the demographic and clinical characteristics of the study sample, and Table 2 summarizes the cognitive factors.

Hip Disability Outcomes at Three Months Postsurgery

Figure 1 shows the distribution of the dependent variable. The HOOS at three months postsurgery had a mean of 79.9 and an SD of 14.4; it was not normally distributed, with significant kurtosis (1.67) and skewness (−1.11). The transformation would likely have created better model fits, but for the sake of a more straightforward interpretation of results, we decided to work with the HOOS's original distribution.

Results of LASSO Models

Machine-learning methods can result in multiple effective predictive models. No single model is necessarily the correct one, and the inclusion/exclusion of other predictors can affect a given

Table 1
Descriptive Statistics of Study Sample at Baseline^a (N = 160).

Variable	Mean	SD
Age	66.7	9.7
Range	34–84	
Comorbidities ^b , out of 15 presented	2.2	1.2
Range	0–5	
BMI	28.3	4.7
Range	15–40	
HOOS ADL at Baseline (higher is better)	43.3	17.2
Range	0–90	
Follow-up Time in Days	126	119
Range	33–458	
Exercise (>15 min; number of days per week, range 0–7)		
Strenuous	0.8	1.6
Moderate	1.9	2.3
Mild	3.9	2.6
Frequency		%
Surgical approach		
Anterolateral	47	29%
Direct Lateral	74	46%
DAA	35	22%
Missing	4	3%
Specific comorbidities ^b		
Arthritis	146	91%
Asthma	12	8%
Back pain	87	54%
Cancer - now or in the past	17	11%
Depression	26	16%
Diabetes	10	6%
Heart disease	14	9%
High blood pressure	51	32%
Insomnia	35	22%
Kidney disease	4	3%
Liver disease	0	0%
Lung disease	1	1%
Stroke	4	3%
Ulcer or stomach disease	4	3%
Other	30	19%
Smoking status		
Never smoked	79	49%
Used to smoke	68	43%
Currently smoke	11	7%
Missing	2	1%
Gender		
Male	89	56%
Female	71	44%
Marital status		
Married	110	69%
Widowed	9	6%
Living with significant other	10	6%
Single (never married)	13	8%
Divorced/Separated	15	9%
Civil Union/Domestic Partner	3	2%
Race		
First Nations/Native Canadian	2	1%
Asian	3	2%
Black	2	1%
White	148	93%
Missing	5	3%
Hispanic ethnicity		
Yes	2	1%
No	112	70%
Missing	46	29%
Work status		
Currently working	62	39%
On leave of absence	5	3%
Retired (not due to ill health)	76	48%
Disabled and/or retired because of ill health	4	3%
Homemaker	1	1%
Unemployed	2	1%
Other	10	6%
Hours worked per week		
Does not apply	73	46%
Less than 20 h	22	14%
20–29 h	9	6%

Table 1 (continued)

Variable	Mean	SD
30–39 h	17	11%
40 or more hours	37	23%
Missing	2	1%
Workers' compensation for current condition (on or planning to apply)		
Yes	1	1%
No	154	96%
Missing	5	3%
Considering legal action for current condition		
Yes	2	1%
No	149	93%
Missing	9	6%
Level of education		
Less than high school	7	4%
Graduated from high school or earned GED	14	9%
Some college or technical school	23	14%
Graduated from college	46	29%
Postgraduate school or degree	69	43%
Missing	1	1%

^a Data reflect baseline values for all variables except follow-up time in days.^b For these topics, a nonresponse was counted as the absence of the event in question (no disease, no use of the medication, etc.) except for in imputation as described in the text.

predictor's strength and even its sign. Even so, LASSO notably failed to include several variables that are often considered important in predictive modeling in surgery research. Smoking status was not found to be an effective predictor, nor was age, level of education, or frequency of any except strenuous exercise.

Table 3 describes A) the regression model resulting from the multiple stages of LASSO analyses described earlier, and B) the final model, with the addition of the Surgical Approach by forced entry. These models explained 29% and 31%, respectively, of the variance in HOOS in the full data set, with the latter usually decreasing on cross-validation (mean = 10%, SD = 7%, range: 0% to 38%). Results from bootstrapping coefficients and cross-validating R^2 for the final model appear in Table 3 as well. The reliability of coefficients was often strong as judged by the ratio of b to bootstrapped standard error (SD_{boot}). For 14 of 20 coefficients, this ratio exceeded the oft-used yardstick of 1.96, and for many, it exceeded 3.

Regression Diagnostics

Diagnostic plots revealed at most moderate difficulties with the final model. Partial plots showed relationships to be consistently linear and portrayed no influential points that seriously affected estimates. Residuals followed a somewhat skewed distribution but were only slightly heteroscedastic. They did not change substantially in mean at different levels of predicted HOOS, though they did vary according to levels of HOOS itself. Figure 2 shows a scatter plot of predicted vs observed HOOS, revealing fairly good adherence of the data to the linear fit line, with the smoothed line indicating no meaningful nonlinearity.

Final Predictive Model

Figure 3 shows the standardized beta estimates for the predictors retained in the final model that had at least small effect sizes. The final model's three strongest predictors were all indicators of cognitive-appraisal processes. Frequent thoughts about accomplishing new goals at work and about comparing oneself to others whose health does not limit them were linked with distinctly worse outcomes ($\beta = -0.34$ and -0.26 , respectively). The two strongest remaining predictors, BMI and number of comorbidities, had unsurprising negative coefficients ($\beta = -0.17$ and -0.19 , respectively). The surgical approach contributed modestly to the model, accounting for one additional point of adjusted R^2 . The

Table 2
Descriptive Statistics of Appraisal Items at Baseline (N = 160).

Quality-of-Life Definition Item (Frequency of Thoughts: 1 = Always to 5 = Never) ^a	N	Mean	SD
Maintaining a positive outlook, even when things are going badly	158	2.33	1.19
Achieving a calmer, more peaceful or healthier lifestyle	156	2.57	1.15
Solving problems that you have with health care or the health care system	153	3.14	1.32
Impressions and assumptions that others have about you because of your health	156	3.65	1.18
How you compare to others whose health does not limit them	154	3.56	1.19
Being free of money problems	154	3.75	1.32
Staying active and productive	158	2.28	1.19
Resolving problems in your living situation	155	3.21	1.41
Preparing your family for the ups and downs of your health condition	156	3.15	1.25
Settling conflicts with people in your life	152	3.65	1.24
Learning to accept yourself as you are	156	3.33	1.42
Spending more time with your family before your health worsens	155	3.57	1.33
Remaining independent and being able to get around on your own	156	2.42	1.39
Being rid of obligations and responsibilities	155	3.79	1.20
Trying not to complain about your health to others	157	3.05	1.36
Increasing your travel for leisure or for visiting with people	155	3.15	1.39

^a Published with permission from DeltaQuest Foundation, the owner of this copyrighted measure.

frequency of thoughts about increasing volunteer work showed the strongest positive association ($\beta = 0.22$), followed by focusing on others' impressions and assumptions about oneself, trying not to complain to others, and family-related goals ($\beta = 0.16, 0.13, 0.12$, and 0.12 , respectively). Being married and currently working had modest, positive links with HOOS, while high blood pressure evinced a counterintuitively positive relationship ($\beta = 0.14, 0.17$, and 0.13 , respectively). Other predictors chosen through LASSO were assigned coefficients shrunk to the point where they played minor roles in this solution.

Discussion

Despite having PROMs data available to them, clinicians continue to have limited ability to identify patients at risk of having a mismatch in outcome following THA. As such, accurate predictive models are needed to help guide shared decision-making and personalize a patient's expected surgical experience. While generalized linear models have been used in the past, new machine-learning techniques are often able to outperform traditional linear statistical models [50–53]. With a supervised machine-learning approach, including the use of LASSO for selecting

variables, we were able to identify multiple patient-specific factors that could help explain why similar patients experience different outcomes despite having the exact same pathology, surgeon, and operation.

A large proportion of previous work in outcome-based research has focused on identifying patient demographics and clinical variables that influence the surgical outcome. Research has shown that this is just one small piece of the puzzle, with the patient outcome being significantly linked to psychosocial health, patient perspective, and the process of how a patient thinks about his/her symptoms, values, and functional health over time (ie, cognitive appraisal) [18,19,30,54,55]. The importance of this is highlighted by the results of our model in which the final three strongest predictors of outcome were all indicators of cognitive-appraisal processes. Frequent thoughts about accomplishing new goals at work and about comparing oneself to other healthier individuals were consistently linked with worse outcomes. On the other hand, patients with a strong social support system, as measured by living with a partner, being focused on family-related goals, and being motivated to help others in their community, could be predicted to have a better result.

The key contributing patient demographic and clinical variables in the model are in line with previous publications in the total knee arthroplasty population [18,20]. Patients with an increased BMI and a greater number of medical comorbidities were at risk of comparably worse postoperative function. Similarly, in the spine population, it is well established that a patient's employment status is an important predictor of outcome [56–58]. This can often be attributed to a variety of factors, including the motivation to return to work, positive reinforcement from social networks, and the physical benefits that daily work entails. Accordingly, patient employment was a positive predictive factor in this study.

In contrast, our findings suggest worse outcomes for people receiving a direct anterior surgical approach. There has been much debate for decades regarding the optimal surgical approach for THA [59–61]. Recently, direct anterior approach (DAA) has experienced significant uptake as an approach with performance of anterior THAs in the United States estimated to have increased from 12% in 2009 to 40% in 2018 [62,63]. Advocates for DAA cite a muscle-sparing approach, shorter time of recovery, and faster rehabilitation [64]. Despite increased utilization of DAA, the success of the approach has been directly linked to surgeon experience and a significant learning curve of 50–100 cases [65,66]. If the approach is not mastered, it is associated with an increased risk of revision surgery [67]. In our cohort, patients undergoing DAA had a poorer

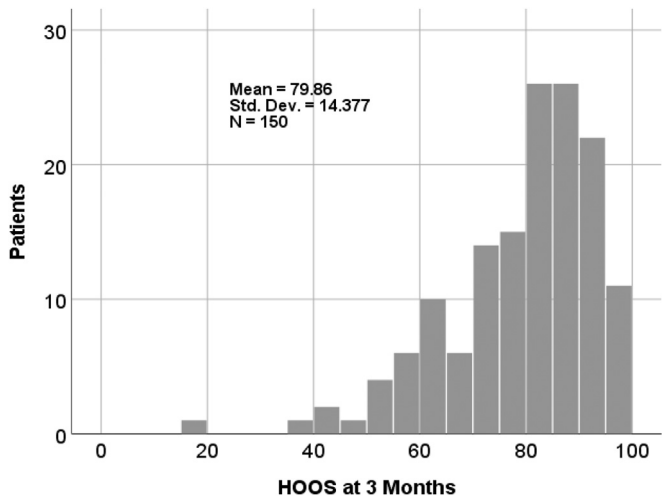


Fig. 1. Histogram showing the distribution of the outcome variable used in modeling. The outcome variable HOOS was negatively skewed. It included responses collected 1.5–6 months postsurgery, with a mean of 2.8 months.

Table 3
Regression models.

Predictor	A. Predictors Chosen via LASSO (N = 128)		B. Final Model, with Surgical Approach (N = 126)						
	<i>b</i>	SE	<i>b</i>	SB	SB <i>boa</i>	<i>b</i> /SD _{boot}	Zero-order <i>r</i>	Tol	Beta
Surgical Approach: Direct Lateral (reference category)	NA	NA	NA	NA	NA	NA	−0.07		NA
Surgical Approach: Anterolateral	NA	NA	1.2	2.6	0.9	1.3	0.11	0.67	0.04
Surgical Approach: Direct Anterior	NA	NA	−5.6	3.5	1.4	4.1	−0.04		−0.15
Married (Or Living with a Partner)	3.1	2.6	4.2	2.7	0.9	4.8	0.13	0.83	0.14
Currently Working	4.2	2.4	4.7	2.4	0.9	5.0	0.18	0.83	0.17
Number of Comorbidities	−1.6	1.4	−2.1	1.4	0.5	3.9	−0.19	0.37	−0.19
Body Mass Index	−0.5	0.3	−0.5	0.3	0.1	4.5	−0.14	0.77	−0.17
Female	−2.2	2.6	−2.4	2.6	0.9	2.8	−0.10	0.71	−0.09
Strenuous Exercise (Days per Week)	0.1	0.5	0.2	0.5	0.1	1.5	0.12	0.77	0.04
Depression	−1.8	3.8	−2.5	3.8	1.7	1.5	−0.07	0.59	−0.07
High Blood Pressure	2.5	2.9	3.7	3.0	1.1	3.4	−0.01	0.59	0.13
Insomnia	−0.1	3.2	1.0	3.3	1.2	0.8	−0.14	0.63	0.03
Cognitive Appraisal: Frequency of Thoughts About....									
Achieving a calmer, more peaceful or healthier lifestyle	−0.2	1.3	0.0	1.3	0.5	0.0	0.11	0.48	0.00
Impressions and assumptions that others have about you because of your health	1.5	1.2	1.7	1.2	0.5	3.6	0.19	0.53	0.16
How you compare to others whose health does not limit them	−2.6	1.2	−2.8	1.2	0.5	5.5	0.11	0.48	−0.26
Increasing your volunteer work to help others in your community	2.4	1.3	2.4	1.3	0.5	4.7	0.23	0.43	0.22
Growing spiritually	0.1	1.1	0.0	1.1	0.4	0.1	0.16	0.53	0.00
Resolving problems in your living situation	1.3	1.1	1.1	1.1	0.4	3.0	0.26	0.48	0.12
Spending more time with your family before your health worsens	1.3	1.2	1.2	1.2	0.4	2.8	0.29	0.43	0.12
Trying not to complain about your health to others	1.2	1.0	1.3	1.0	0.4	3.3	0.23	0.63	0.13
Increasing your travel for leisure or for visiting with people	0.0	1.2	0.2	1.2	0.4	0.5	0.11	0.40	0.02
Accomplishing new goals at work	−2.8	0.9	−2.9	0.9	0.3	8.6	−0.08	0.53	−0.34
R ²	0.29		0.31						
Adjusted R ²	0.17		0.18						

Models have been created for a predictive, not explanatory, purpose. Coefficients cannot necessarily be given substantive interpretations because of underlying collinearity, suppressor effects, and/or mediation effects.

Conditional formatting to indicate effect size with bold font indicating a negative parameter coefficient and italicized font reflecting a positive coefficient parameter. Font that is in bold and italicized represents the factor with the greatest effect size.

b, coefficient; SE, standard error; SDboot, bootstrapped standard deviation as an alternative to SE; VIF, variance inflation factor; Tol, tolerance; Beta, standardized *b*.

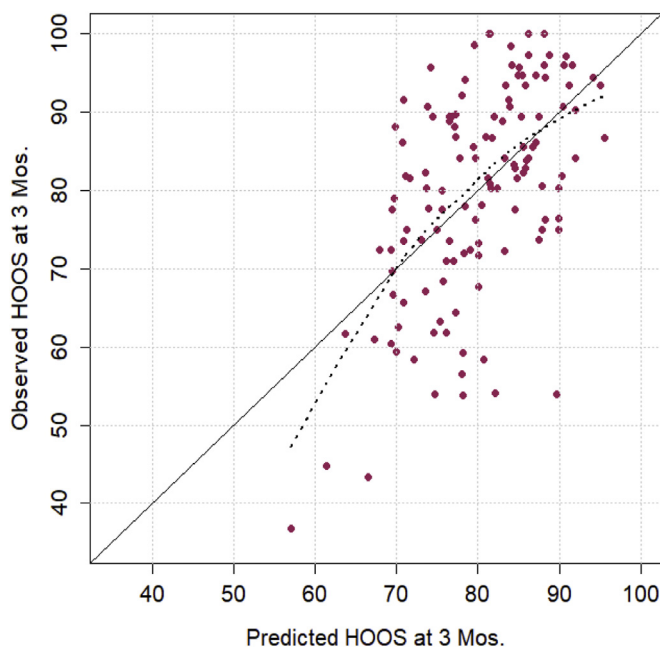


Fig. 2. Predicted vs Observed HOOS. Scatter plot of Observed vs Predicted HOOS. Mean residuals (prediction errors) are roughly the same across values of Predicted HOOS. Dotted line = Locally Weighted Scatter plot Smoother (LOWESS). As Observed HOOS increases, HOOS predicted by the model increases as well, and the relationship is quite linear.

HOOS score at 3 months follow-up when compared to the direct lateral approach. While our findings would appear to contradict the proposed early benefit of DAA, it is important to note that one of our two DAA surgeons was within the aforementioned learning curve period during patient enrollment. The anterolateral approach had no impact in predicting postoperative HOOS when compared to the reference direct lateral approach.

The value of this model should be judged based on a few factors. First, the patient population described here is representative of the typical demographic of patients undergoing routine primary arthroplasty, making it generalizable to most surgeons and patients considering THA. The surgical success rate of our intervention similarly matches the standard of care available in the literature [3,10,68], with significant preoperative to postoperative change in HOOS, from a mean of 43 at baseline assessment and 80 at 3 months postoperatively.

A few models in the total knee arthroplasty population do exist, allowing comparisons to be made to our findings [20,37]. The predictive accuracy of a model is described by its R^2 value, that is, how well the identified independent variables can explain the outcome. An ideal predictive tool should explain at least 30% of the variability of the outcome, as in the Framingham Heart Study score [69]. Our R^2 of 31% and adjusted R^2 of 18% fits within this range of predictive accuracy and likely could be improved with transformation of the HOOS data. Strong statistical predictors were found, as detailed above, allowing conclusions to be drawn about how changes in the predictor values are associated with changes in the outcome.

While a machine-learning approach can be useful in predicting outcomes, it should not be confused with causal inference (ie, this

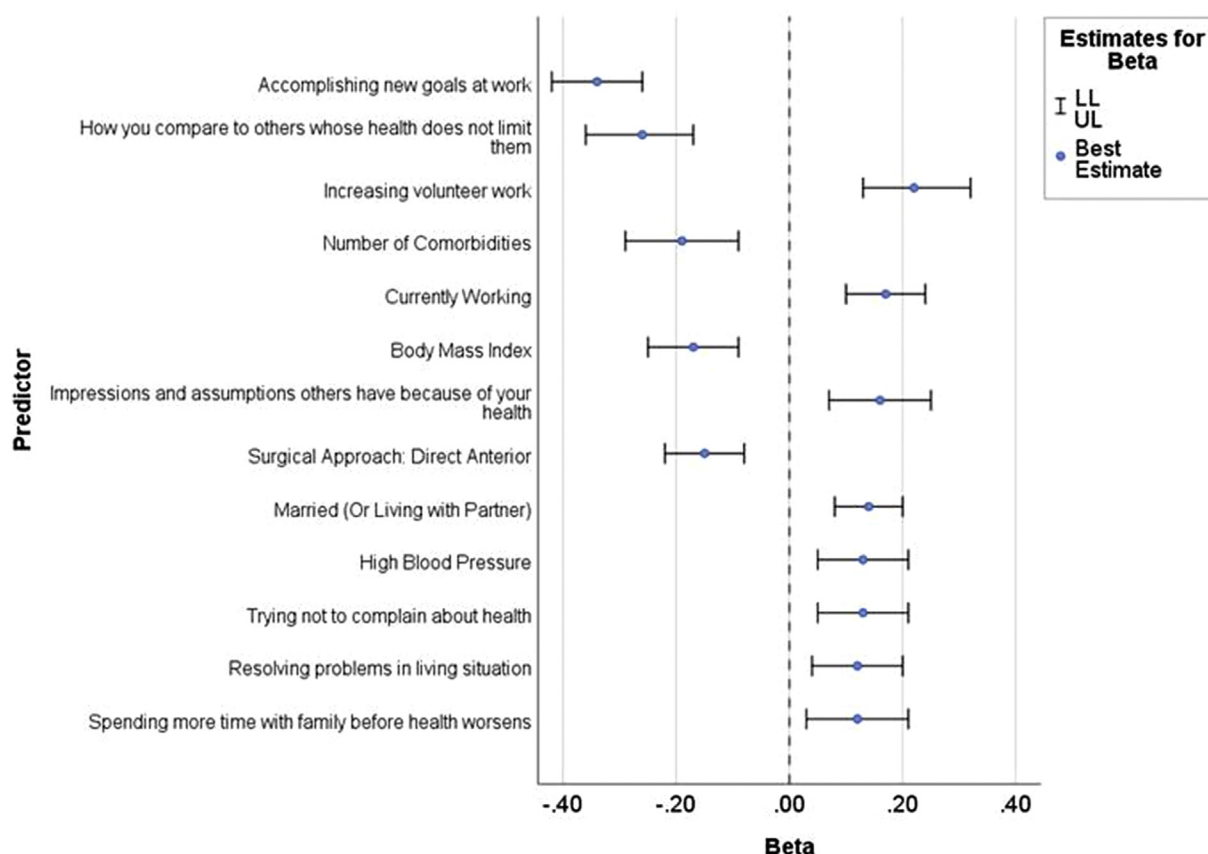


Fig. 3. Coefficient Plot with Bootstrapped Estimates for Beta. Predictors appearing higher in the plot are those with stronger predictive relationships with NRS at ~10 Months. Beta = standardized regression coefficient. At a Beta of 0.4, an increase of 1 SD on the predictor is associated with an increase of 0.4 SD on the outcome. Upper and lower limits for Beta are based on 1000 iterations of bootstrapping.

variable causes this outcome). Utilizing prediction to understand the variables that are linked with the outcome allows for an improved shared decision-making discussion. They provide important context for supporting individuals at risk, and they can facilitate more informed patient selection for THA.

Being able to identify patients who may be at risk of a poor outcome without any surgical-related cause is important at an early stage, whether it be at the preoperative patient visit or early postoperative follow-up. This study is performed with a follow-up of 3 months, which is useful with this goal in mind. We recognize that this short follow-up is a limitation in our study. While many postoperative THA patients are doing very well at this point, others are continuing to recover, with further improvement expected. However, being able to identify patients at risk of a poor outcome at earlier follow-up may help predict those at risk of similar outcomes 6–12 months postoperatively. As our patient cohort is prospective in nature, follow-up is ongoing and will be available in the future for analysis. Similarly, patient recruitment is ongoing, with the potential for a larger patient cohort in the future and a potentially more robust model. Finally, as this model is specific to THA, it may not be generalizable to other orthopedic populations.

In conclusion, we have developed a machine-learning model that is able to identify independent variables predictive of THA functional outcome. This has the potential to improve patient care by informing the shared decision-making process of a patient's likely outcome from THA and may also provide an opportunity to address modifiable risk factors that would set the patient up for a higher likelihood of success. We recommend that surgeons

continue utilizing the surgical approach to the hip that they are most comfortable with but should also be aware of the steep learning curve associated with DA arthroplasty. Patients with an increased BMI, multiple medical comorbidities, and limited social support should be counseled of their risk of a poorer functional outcome following THA compared to their peers. Appropriate consultations to a dietician, internal medicine physician, or social worker should be arranged preoperatively to help address these risks. As the factors most predictive of outcome were cognitive appraisal processes, consideration should be given to administering one of the available appraisal measures, such as the Brief Appraisal Inventory [42] or the QOL Appraisal Profile_{v2} [70], at the initial preoperative visit. If risk factors are identified via the questionnaire, if deemed appropriate, a consultation can be made to a cognitive-behavioral therapist in the perioperative period. As our understanding of surgical outcomes in arthroplasty continues to evolve beyond that of just physical and objective data, incorporating more measures of patient perspective, and cognitive processes into future work appears to be imperative. These measures may help us better understand the frustrating patient-surgeon, observed vs expected mismatch in the outcome that can occasionally be seen in patients undergoing THA.

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